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RED ID: 823343580

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Lab 1 Report

What does your program do?

My program outputs my name and RED ID in morse code through an LED on the ATMEGA328PB Microcontroller.

What variables/registers does your program use?

My program uses register the B5

What is the main algorithm/logic of your program?

The main algorithm of the program is to blink the LED according to the Morse code representation of my name "Abdulmohsen" and my RED ID which is 823343580. This is achieved by defining functions for dots, dashes, and each letter of the name. The main function then calls these functions in order to blink out the desired Morse code according to the international Morse code.

What external hardware connections did you make?

Only a connection to Atmel studio board

C CODE

```
#define F_CPU 16000000UL
#include <avr/io.h>
#include <util/delay.h>

void dot(void) {
   PORTB |= (1 << PORTB5);
   _delay_ms(200);
   PORTB &= ~(1 << PORTB5);
   _delay_ms(200);
}

void dash(void) {
   PORTB |= (1 << PORTB5);
   _delay_ms(600);
   PORTB &= ~(1 << PORTB5);
   _delay_ms(200);
}

void letter_A(void) { dot(); dash(); _delay_ms(400); }
void letter_B(void) { dash(); dot(); dot(); dot(); _delay_ms(400); }
void letter_C(void) { dash(); dot(); dash(); dot(); _delay_ms(400); }
void letter_D(void) { dash(); dot(); dot(); _delay_ms(400); }
void letter_E(void) { dot(); _delay_ms(400); }
void letter_F(void) { dot(); dot(); dash(); dot(); _delay_ms(400); }
void letter_G(void) { dash(); dash(); dot(); _delay_ms(400); }
void letter_H(void) { dot(); dot(); dot(); dot(); _delay_ms(400); }
void letter_I(void) { dot(); dot(); _delay_ms(400); }
void letter_J(void) { dot(); dash(); dash(); dash(); _delay_ms(400); }
void letter_K(void) { dash(); dot(); dash(); _delay_ms(400); }
void letter_L(void) { dot(); dash(); dot(); dot(); _delay_ms(400); }
void letter_M(void) { dash(); dash(); _delay_ms(400); }
void letter_N(void) { dash(); dot(); _delay_ms(400); }
void letter_O(void) { dash(); dash(); dash(); _delay_ms(400); }
void letter_P(void) { dot(); dash(); dash(); dot(); _delay_ms(400); }
void letter_Q(void) { dash(); dash(); dot(); dash(); _delay_ms(400); }
void letter_R(void) { dot(); dash(); dot(); _delay_ms(400); }
void letter_S(void) { dot(); dot(); dot(); _delay_ms(400); }
void letter_T(void) { dash(); _delay_ms(400); }
void letter_U(void) { dot(); dot(); dash(); _delay_ms(400); }
void letter_V(void) { dot(); dot(); dot(); dash(); _delay_ms(400); }
void letter_W(void) { dot(); dash(); dash(); _delay_ms(400); }
```

```
void letter_X(void) { dash(); dot(); dot(); dash(); _delay_ms(400); }
void letter_Y(void) { dash(); dot(); dash(); dash(); _delay_ms(400); }
void letter_Z(void) { dash(); dash(); dot(); dot(); _delay_ms(400); }


void number_0(void) { dash(); dash(); dash(); dash(); dash(); _delay_ms(400); }
void number_1(void) { dot(); dash(); dash(); dash(); dash(); _delay_ms(400); }
void number_2(void) { dot(); dot(); dash(); dash(); dash(); _delay_ms(400); }
void number_3(void) { dot(); dot(); dot(); dash(); dash(); _delay_ms(400); }
void number_4(void) { dot(); dot(); dot(); dot(); dash(); _delay_ms(400); }
void number_5(void) { dot(); dot(); dot(); dot(); dot(); _delay_ms(400); }
void number_6(void) { dash(); dot(); dot(); dot(); dot(); _delay_ms(400); }
void number_7(void) { dash(); dash(); dot(); dot(); dot(); _delay_ms(400); }
void number_8(void) { dash(); dash(); dash(); dot(); dot(); _delay_ms(400); }
void number_9(void) { dash(); dash(); dash(); dash(); dot(); _delay_ms(400); }

void space(void) { _delay_ms(1200); }

int main(void) {
   /* Replace with your application code */
   DDRB |= (1 << DDB5);  // Set port bit B5 in the data direction register to 1 (configure
as output)

   while (1) {
      letter_A(); letter_B(); letter_D(); letter_U(); letter_L(); letter_M(); letter_O();
letter_H(); letter_S(); letter_E(); letter_N();
      space();


      number_8(); number_2(); number_3(); number_3(); number_4(); number_3();
number_5(); number_8(); number_0();
      space();
   }
}
```